Problem Set 6: MLE

DS5030 Understanding Uncertainty

1. Let $\ell(\lambda) = a \log \lambda - b\lambda$, with $a, b > 0$. Find the value of λ that maximizes ℓ by setting $\ell'(\lambda) = 0$. Check that the second derivative is negative.

2. Let $X_1, \ldots, X_n$ be i.i.d. $N(\mu, \sigma^2)$ with observed values $x_1, \ldots, x_n$. Show your steps.

   a. Write the log-likelihood $\ell(\mu, \sigma^2)$. Then maximize it to find $\hat{\mu}$ and $\widehat{\sigma^2}$ (Hint: we did this in class.)

   b. For $x = (3.0,\ 5.0,\ 9.0)$, compute $\hat{\mu}$ and $\widehat{\sigma^2}$.

   c. With $\sigma = 2$, $n = 25$, and $\bar{x} = 10$, compute a 95% confidence interval for μ and a 95% prediction interval for one new observation. Use $z = 1.96$. Which is wider, and why?

3. Let y be an $n \times 1$ vector with $y \sim N(X\beta, \sigma^2 I)$, where X is $n \times p$: n observations of p predictors.

   a. Write the log-likelihood $\ell(\beta, \sigma^2)$ in matrix form.

   b. Show that maximizing ℓ over β is the same as minimizing $(y - X\beta)^\top (y - X\beta)$, and that $\hat{\beta} = (X^\top X)^{-1} X^\top y$.

   c. Take $n = 3$, $y = (2, 3, 5)^\top$, and

$$X = \begin{pmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{pmatrix}.$$

   Compute $\hat{\beta}$. *Hint:* for a 2×2 matrix, $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \dfrac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.